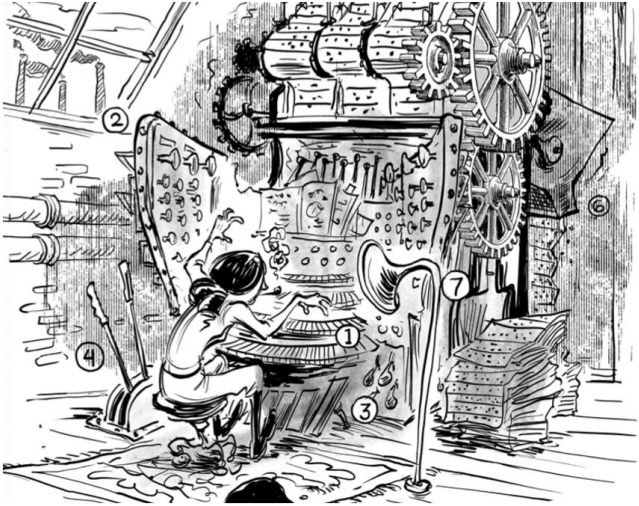
COM2020 Team Project

Lecture 2: Project planning & management

Prof. Solomon Oyelere
Department of Computer Science
s.oyelere@exeter.ac.uk

Vision



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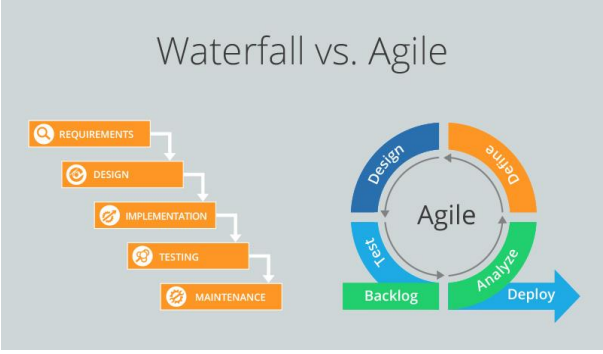
Team Project

Teams apply technical skills in software development and data analysis while managing project planning, teamwork, and communication.

Software Project Development Schemes

Software project development may be organised into software development schemes including:

1. waterfall development;
2. agile development.



Waterfall Development

Waterfall development carries out the software development phases strictly one after the other.

Advantages

Waterfall development works well when requirements are fixed, technology is well-understood, and the project is short or done in isolation.

Disadvantages

Waterfall development works poorly when requirements appear unexpectedly, technology changes, and the project is unbounded.

Agile Development

Agile development carries out the software development phases in an iterative, incremental way.

Advantages

Agile development works well when requirements emerge as a result of regular customer consultation, and there is no difficulty with regular iterative updates.

Disadvantages

Agile development works poorly when regular customer consultation is difficult, organisational structure is restrictive, and scope-creep is problematic.

Scrum

A Scrum team is a self-organising, **cross-functional team** consisting of a Product Owner, a Scrum Master, and a Development Team, working in **short sprints**

Kanban

A Kanban team is a self-organising, cross-functional Development Team, working in **one continuous sprint**.

Scrum agile methodology

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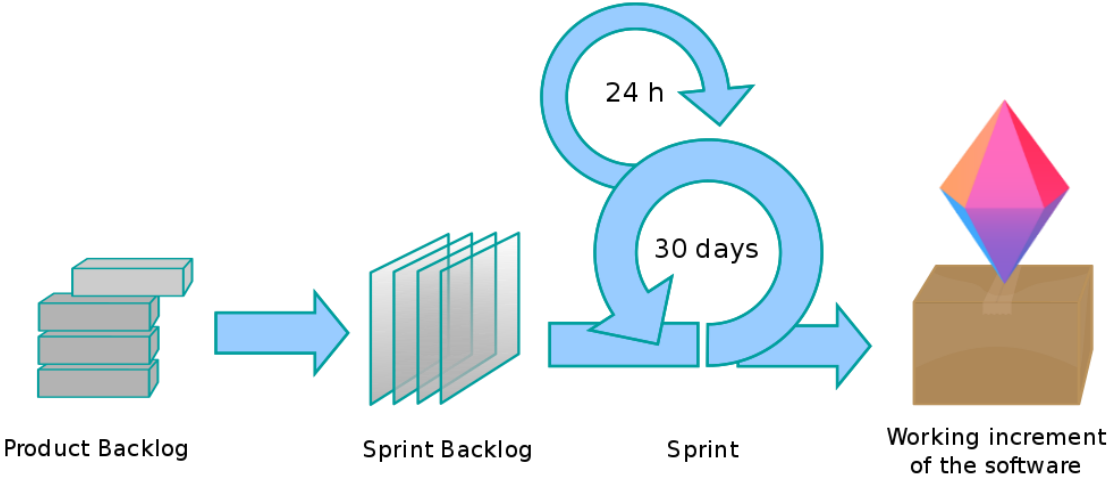
Batch-like iterative production



Scrum Events (ceremonies)

Central to the operation of Scrum are a number of **non-optional**, **time-boxed** prescribed events:

- the sprint;
- sprint planning;
- the (week) daily scrum;
- the sprint review;
- the sprint retrospective

The Sprint

A sprint is time-boxed event (maximum one month), during which a **potentially releasable product** increment is created.

Sprint Planning

Sprint planning is time-boxed event (maximum eight hours), during which the team establishes:

- **what work is to be done** (the **end**);
- **how the work is to be done** (the **means**).

The (weekly) Daily Scrum

The (weekly) **daily scrum** is a time-boxed event (maximum fifteen minutes), during which the team **synchronise activities** and create a **plan for the next 24 hours**.

Typically around 15 minutes

Stand-up meeting

The (weekly) Daily Scrum

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During the meeting, each team member explains:

- what they **did yesterday** to help reach the sprint goal;
- what they **will do today** to help reach the sprint goal;
- what **impediments** they see to reaching the sprint goal.

The Sprint Review

The **sprint review** is a time-boxed event (maximum four hours), during which the team consider **the product**:
what work was done in the current sprint;
what work is to be done in the next sprint.

Emphasis on ‘Product’ and PO

The Sprint Retrospective

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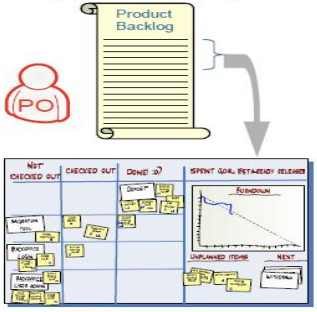
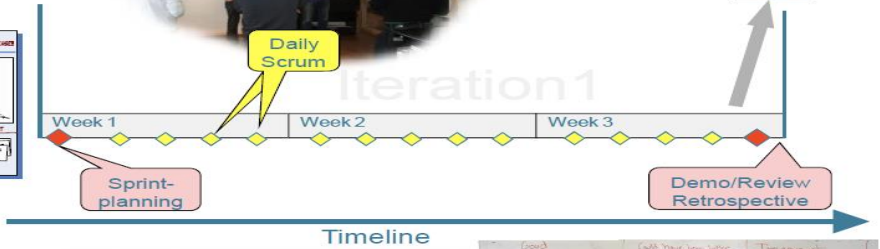
The **sprint retrospective** is a time-boxed event (maximum three hours), during which the team consider **the process**:
how did people/processes/tools work in the current sprint;
how might people/processes/tools work in the next sprint.

Emphasis on ‘Process’ and SM

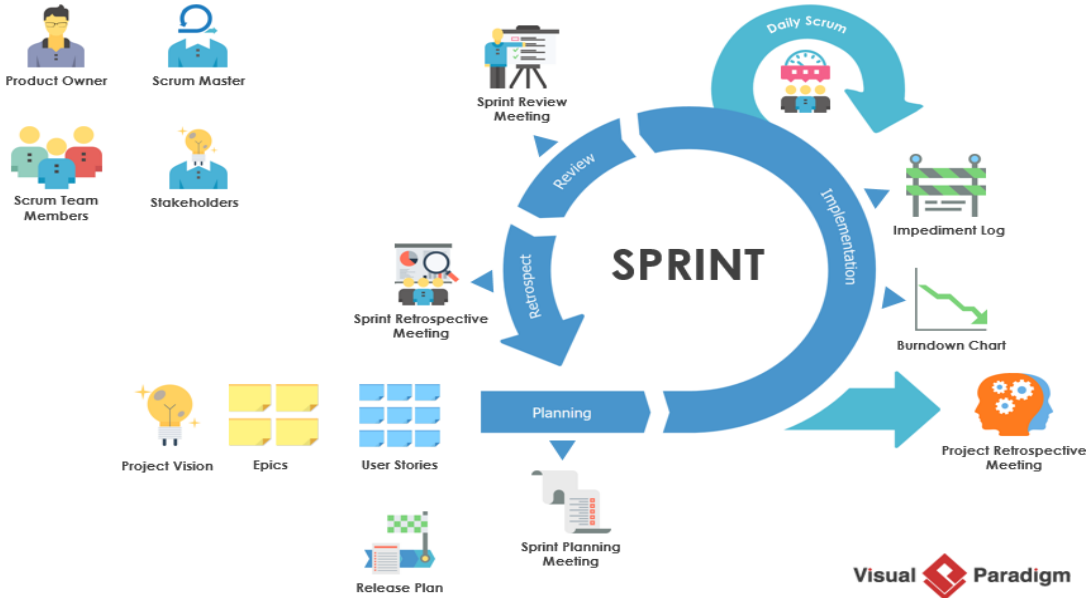
Typical sprint

Principle #8:
Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.





The Agile – Scrum Framework



Populating the Backlog

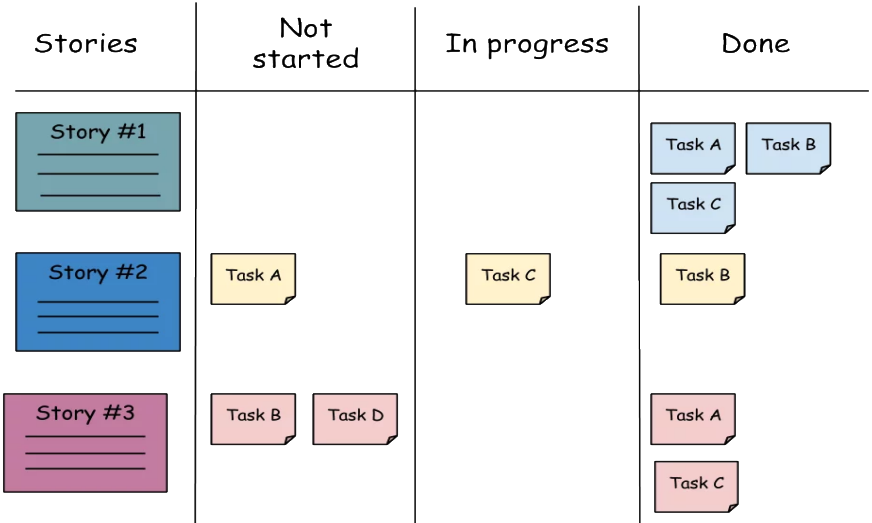
MoSCoW – Must have, Should have, Could have, Won't have

Release cycle planning

Project Backlog vs **sprint backlog**

- Epics
- User stories
- Priority
- Story points

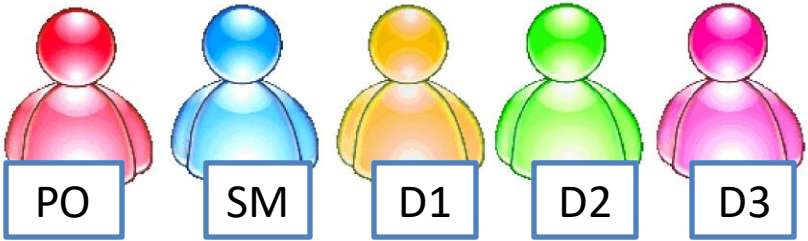
Scrum Board


Stories
Not started
In progress
Done
Story #1
Task A
Task B
Task C
Story #2
Task A
Task C
Task B
Story #3
Task B
Task D
Task A
Task C

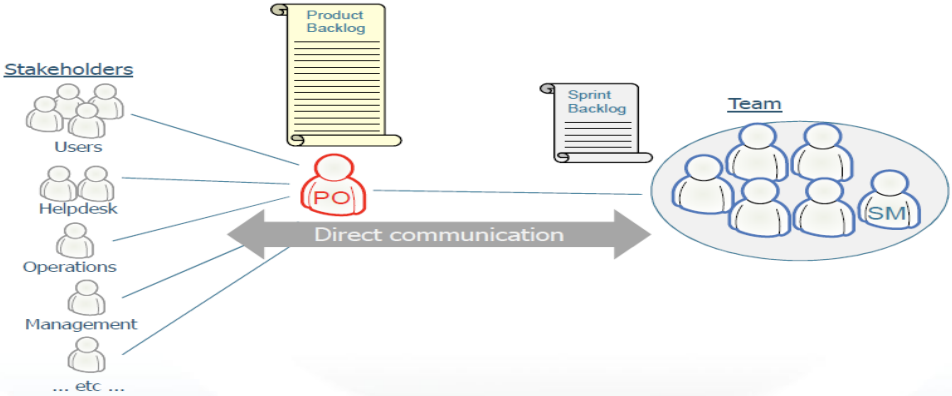
The Scrum Team

The Scrum team is a **self-organising**, **cross-functional** team consisting of:

1. the **Product Owner (PO)**;
2. the **Scrum Master (SM)**;
3. the **Development Team (D1, ...)**.

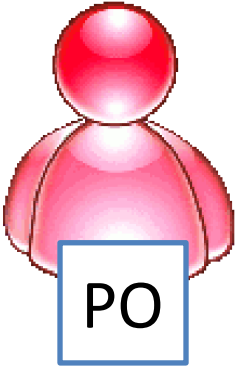


Scrum communication structure



The Product Owner

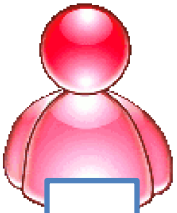
The **product owner** is responsible for **product success** — for **creating a product that does a great job** for customers that meets its business goals.



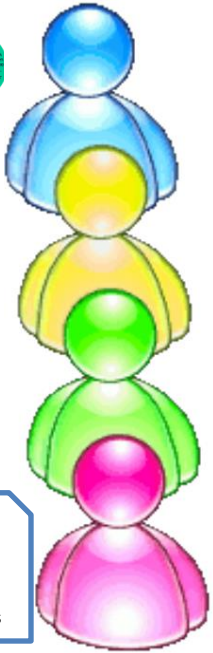
The Product Owner

The product owner **provides the interface** between the Scrum team, customers and senior management

Customer



PO



Senior manager

Product delivery documents

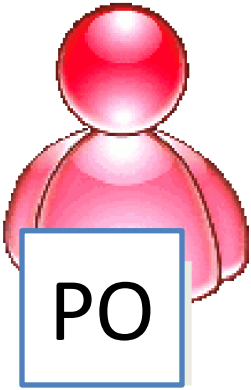
Research and decision making documents

Why is it important to have a single product owner?

What characteristics might you expect of a product owner?

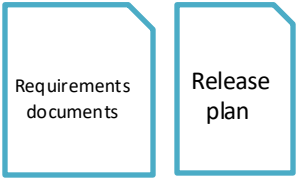
The Product Owner

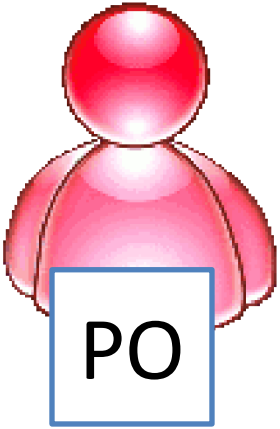
The product owner serves the Scrum team by **providing vision and strategy**, describing where the product is heading, and communicating its value proposition and business goals.



The Product Owner

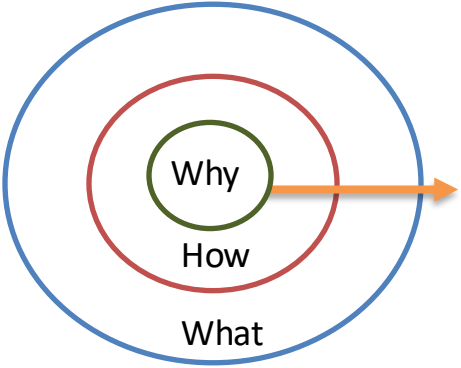
The product owner serves the Scrum team by **providing product details** making prioritisation calls between them, often on the basis of customer feedback.

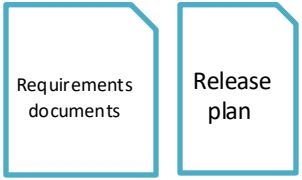




Product owner as leaders

Leaders work outwards - Why > How > What



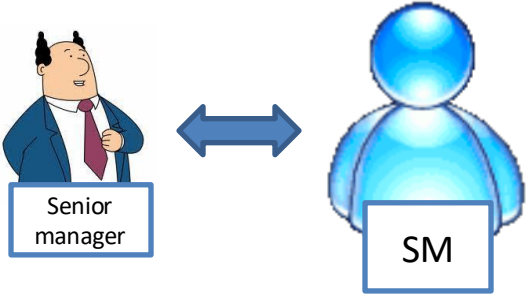


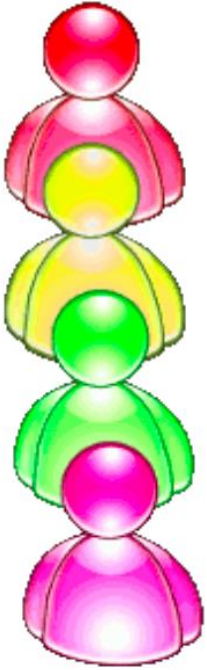
The Scrum Master

The Scrum Master is responsible for **process success** — for helping the development team to use an agile way of working.

The Scrum Master

The Scrum Master provides the interface between the **scrum team and senior management** — Human Resources and other business groups.





The Scrum Master

The Scrum Master serves the Scrum team by **facilitating collaboration**, helping its members to **work effectively on the product backlog**.

- Arrange and chair team meetings
- Make sure collaborations are productive
- Resolve political and technical problems
- Manage and motivate people

The Scrum Master

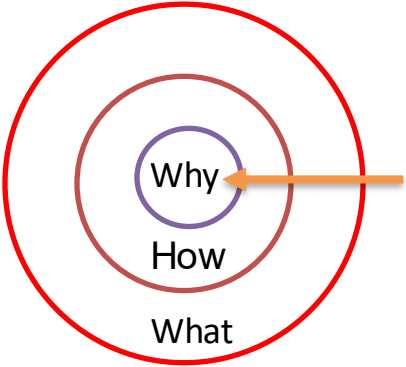
The Scrum Master serves the Scrum Team by **facilitating events** (the **sprint**, **sprint planning**, the **daily scrum**, the **sprint review**, and the **sprint retrospective**).

What characteristics should a Scrum Master have?

Why is it important to distinguish the PO and the SM?

- Organisation – Preparation.
  - Fail to prepare, prepare to fail.
- Time management – punctuality set a good example
- Good person management – mediation understanding what is value to whom
- Problem solving - Technical understanding
- Fairness – Appreciation of technical workload

Scrum Master as a manager



Agile board (with revision history)

Meeting agendas

Meeting minutes

Time sheets

Commit log

Developer document -ation

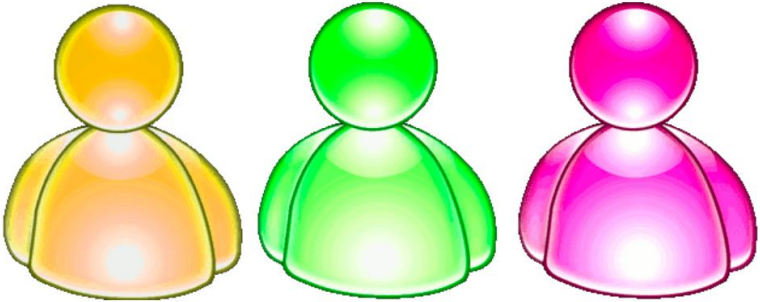
Bug report s

Manager work inwards - What > How > Why
Responsible for process documentation.

What documents might a SM produce?

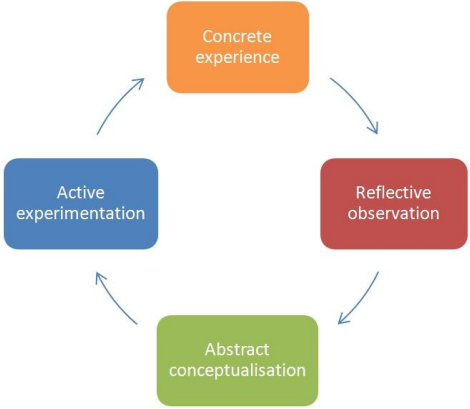
The Development Team

The **Development Team** is responsible for delivering the next (potentially releasable) increment of the product.



The Sprint Retrospective

Look up the Kolb cycle as a framework for reflection

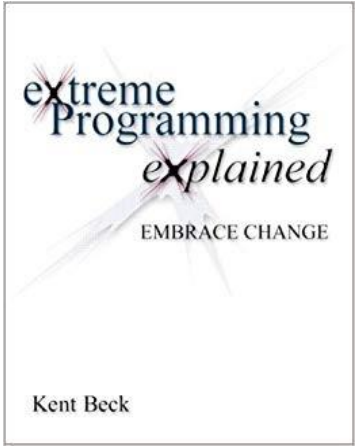


- **Concrete experience**: The initial experience that provokes a reaction
- **Reflective observation**: Reflecting on the experience and noting anything new
- **Abstract conceptualization**: Developing new ideas based on the experience
- **Active experimentation**: Applying the new ideas to new situations

Sustainable Pace

Agile projects proceed at a sustainable pace:

- **Overtime is a symptom of a serious problem** on the project. The XP rule is simple — you **can't work a second week of overtime**. For one week, fine, crank and put in some extra hours. If you come in on Monday and say, "To meet our goals, we'll have to work late again," then you already have a problem than can't be solved by working more hours.



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Planning poker game

An exercise to estimate product backlog for agile teams

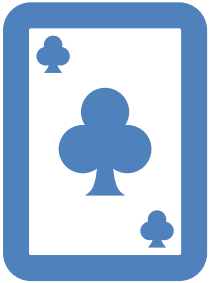
What is Planning Poker?

Planning Poker® is a consensus-based estimating technique. Agile teams around the world use Planning Poker to estimate their product backlogs. Planning Poker can be used with story points, ideal days, or any other estimating unit.

To start a poker planning session, the product owner or customer reads an agile user story or describes a feature to the estimators.

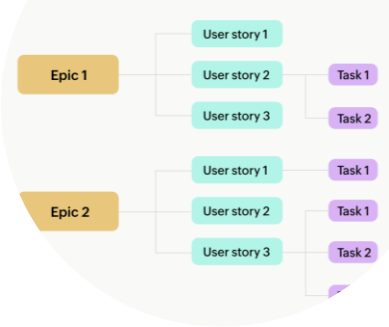
Each estimator is holding a deck of Planning Poker cards with values like 0, 1, 2, 3, 5, 8, 13, 20, 40 and 100, which is the sequence we recommend. The values represent the number of story points, ideal days, or other units in which the team estimates.

Rationale

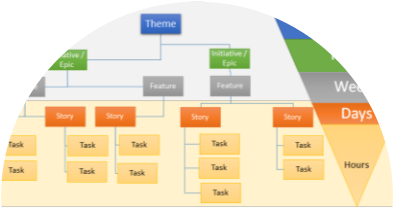


The reason to use planning poker is to avoid the influence of the other participants. If a number is spoken, it can sound like a suggestion and influence the other participants' sizing. Planning poker should force people to think independently and propose their numbers simultaneously. This is accomplished by requiring that all participants show their cards at the same time.

Equipment


Epic 1
User story 1
User story 2
User story 3
Task 1
Task 2
Epic 2
User story 1
User story 2
User story 3
Task 1
Task 1
Task 2

ZOOBOARD
PLANNING POKER CARDS
12 PLAYERS


Theme
Initiative / Epic
Feature
Feature
Story
Story
Story
Story
Task
Days
Hours

Grooming Requirement/Story

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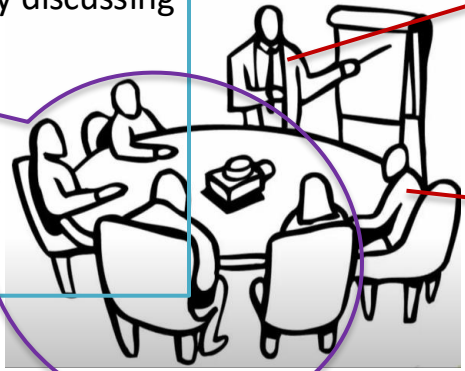
Project Team, mutually discussing
- Amount of work
- Associated risks
- Technical changes
- Dependancies
- Complexities
- And the value



Product owner is explaining all the aspects of the story requirement, dependacies, acceptance criteria and business value

Scrum master facilitating the grooming session

Source: https://youtu.be/BSNLkg_8odQ

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Backlog	Story 13
	Story 12
	Story 11
Groomed	Story 10 [3]
	Story 9 [5]
	Story 8 [3]
In Progressed [Sprint 10]	Story 7 [8]
	Story 6 [5]
Completed	Story 4 [2]
Accepted	Story 1 [5]
	Story 2 [3]

Grooming story 11

Guys, are you clear about the requirement and acceptance criteria. And its Risks dependencies or any other impact?

Can we estimate the story?

Remember, only the development team can estimate by expressing their view points with a number in Fibonacci series (poker card).
They can discuss details with PO, BA or external team member to gather more information to gauge the size better

Source: https://youtu.be/BSNLkq_8qdQ

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Backlog	Story 13
	Story 12
	Story 11
Groomed	Story 10 [3]
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Grooming story 11

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Grooming story 11

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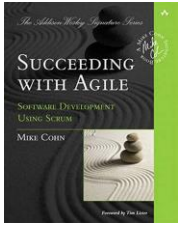
References



M. Cohn, Succeeding With Agile: Software Development Using Scrum, Pearson, 2010, ISBN 978- 0321579362.